

Lesson Overview: Introduction to Big Data

In today's data-driven world, the ability to efficiently handle big data is critical for data engineers. With organisations leveraging data for strategic decisions, proficiency in managing and analysing big data is indispensable, driving the need for scalable, efficient solutions that transform data into actionable insights.

In this lesson, we'll explore the essentials of big data, including its defining characteristics, the challenges of managing vast datasets, and the solutions that modern data storage technologies offer. We'll learn about the distinction between data lakes and warehouses, the importance of choosing the right data storage solutions, and how to design big data strategies that deliver significant business value. We'll build a solid foundation in big data principles and practices as well as an understanding of data persistence when it comes to containerisation, enabling us to navigate and implement effective data management strategies in various industries.



What You'll Learn:

The four Vs of big data

Learn about the volume, velocity, variety, and veracity of big data, which define its characteristics.

Challenges in big data management

Explore the main challenges in managing big data and the technologies that address these issues.

Types of data and data sources

Understand the differences between structured, unstructured, and semi-structured data, and internal versus external data sources.

Ready to dive in?

Reflect:

Think about the different types of data you encounter — social media posts, financial transactions, health data — and consider the challenges that organisations might face when trying to handle and make sense of all this information. How might understanding big data help you contribute to meaningful solutions in your future career?